These tips are good starting points for writing a paper or thesis. Of course it is possible to deviate from these rules (especially for advanced writers) but take this as a solid starting point and be aware of your deviations, if any.

## Major tips

1. Think of the main message of your paper. Some papers have two main messages, but more often it is just one. Try to phrase this message in your title. This will define how you write the rest of the paper: the whole point, and the **only** point, of the rest of the paper is to explain, defend, and demonstrate/answer this main message. Nothing else. For instance, an article could be titled "*Information dissipation as an early-warning signal for the Lehman Brothers collapse in financial time series*" which hints already at the fact that some new analysis is shown to be able to warn against upcoming financial crises and it is tested on real time series data. Do not be generic like "A study of social opinion forming in online networks" and then enumerate a bunch of disconnected results in the Results section: instead, the results you show in your Results section must logically follow each other, building up to eventually prove/demonstrate your main point beyond any doubt.
2. Start then with writing an abstract. It guides the setup of the rest of the paper, so do it carefully. It forces you to think about what is your most important finding, what open questions it answers, and why it is important. As inspiration use www.nature.com/nature/authors/gta/Letter_bold_para.doc and stay as close as possible to it.
3. The Introduction usually follows a similar progression of topics as the abstract, so start with motivation and context and open questions. Then go on to say what you did, followed by optionally a literature review as part of introduction (large literature reviews can instead be put in a separate section), or in reversed order (review first) is also fine, depending on what serves best your logical storyline. Then probably end the introduction with your main finding and its wider implications (usually this is repeated/rephrased/deepened in the end of the conclusion).
    1. Don't try to hold off on giving away your main findings and conclusions as long as possible, but give them already in the abstract (very concise) and Introduction (a bit concise). Your article is not a detective thriller with cliff hangers, it is a scientific work.
4. Let each paragraph convey a single message. To start, phrase each message in a single sentence for yourself and put it down. The ordered list of all these messages, arranged per section header, is the 'skeleton' of your paper which will define your logical structure (storyline). Make sure the next message logically follows the previous: e.g., use a literature overview to highlight how people previously looked at the problem, then follow by how you look at it differently (if appropriate) or where the missing pieces are.

    When writing the entire paragraph, always let the first (or sometimes the second) sentence convey this message. Often you can reuse the sentence you wrote before in the skeleton. Then the rest of the paragraph will explain it more in depth or defend the statement. A person who reads the first sentence (message) and decides to skip the paragraph should not miss anything important.

    The sequence of all first sentences of the paragraphs should result in a reasonable summary of the entire paper.

    Example:

*A system consisting of coupled units can self-organize into a critical transition if a majority of the units suddenly and synchronously change state*[1,2,3]*. For example, in sociology, the actions of a few can induce a collective tipping point of behavior of the larger society*[4,5,6,7,8,9,10,11]*. Epileptic seizures are characterized by the onset of synchronous activity of a large neuronal network*[12,13,14,15,16,17,18]*. In financial markets the participants slowly build up an ever densifying web of mutual dependencies through investments and transactions to hedge risks, which can create unstable 'bubbles'*[19,20,21,22,23]*.*

5. Within each paragraph, also do your best to let the sentences logically follows each other, and connect them as such. Start a next sentence like "The rationale is that...", "Because of this...", "This implies that...", "Namely/However/Nevertheless/Notwithstanding/That is, ...", "... can then ....". Sometimes just a direct sentence without such 'connector' is also fine if the connection to the previous is obvious or it improves the flow of the text.

   An example:

   *The IDL measures to what extent the state of one unit influences the states of other units. As the state of one unit depends on another unit, a fraction of the bits of information that determine its state becomes a reflection of the other unit's state. This creates a certain amount of mutual information among them. A unit can then influence other units in turn, propagating these 'transmitted' bits further into the network. This generates a decaying amount of mutual information between distant units that eventually settles at a constant. The higher the IDL of a system, the larger the distance over which a unit can influence other units, and the better the units are capable of a collective transition to a different state. Because of this we can measure the IDL of systems of coupled units and detect their propensity to a catastrophic change, even in the absence of a predictive model. See* Sections S1 and S2 in the SI *for a more detailed explanation and how it differs from existing indicators.*

   (Notice, e.g., that the second sentence is a 'Namely, ' sentence, but it is omitted [to improve the flow of the text].)

6. Say what you did, not what you did not. E.g., do not start a paragraph with "Since migration is related to many different aspects this study will not be able to cover all." It is true for literally every scientific study ever performed that there was actually more to study, so stating it is moot and uninteresting and distracting.
7. Say what you must, not what you can. This is a very often overlooked best practice, but very important. For instance, a statement like "Generally speaking the Post-Napoleonic, 19th century Netherlands can be divided into a period of repair and a period of industrialisation" only makes sense if you will actually have results relating to this divide, for instance, that the migration patterns are different in these two periods (and why). If no results pertain to this and you just thought it would be nice to tell the reader such general interesting facts and considerations, please think again. (Or write a monograph after your retirement.)
8. Each statement that you make must be backed up. Example: "Migration is highly correlated with geographical income level differences." The easiest way to back up a statement is to add references to previous works which proved/demonstrated this statement. If that does not exist then you will have to prove/demonstrate it yourself. If none of these two are possible, then don't make the

statement. Only other way: make it an assumption.

9. Make statements as precise and to-the-point as you can, but not too much, insofar that it serves the story. So instead of "Migration is connected to social developments", possibly you actually wanted to say: "Migration is highly correlated with regional differences in income levels.", but do not go to far in each sentence and go autistic-style like "The number of individuals who changed their home address from X to Y in a given year will have a Pearson correlation with the income level difference income(Y) - income(X) which is higher than 0.6, when considering the years 1900,...,1999 for the region ABC." (Naturally, your paper on the whole must be reproducible and thus make all these details clear, but not in a single sentence.)
10. Your paper, including the appendices, must be reproducible.
11. Keep your main text for the 'red thread' of your storyline which includes only the most pertinent results and discussion. Use the appendix (or 'supplementary material') to show other supporting results. For instance, a sentence in the main text could be "This result is insensitive to the choice of parameter value *w* (see Appendix A6)." and then in appendix section A6 you show 10 plots for different parameter values, or some more condensed form of it with the parameter value on the x-axis and the quantity of interest on the y-axis. It is quite possible that the appendix is (much) larger than the main text nowadays. For instance, this article has a main text of about 6 pages and an appendix of about 40 pages.
12. Never make references to data 'not shown'; in the digital age there is no valid reason for omitting any substantiation.
13. Make your Figure captions as complete and stand-alone as possible. If your caption is a single sentence then it is likely that it is not complete yet. Think of a reader who only reads your abstract and then reads your figures and figure captions in order. Could this reader understand your research, at least the main points? It is often advisable to also include the main take-away message in the caption to direct the reader's attention, especially if your figure is complex.

    So do not write something like "*The level of activity of a set of neurons under a microscope.*"

    For instance, this could be a good caption:

    *The level of activity of a set of neurons under a microscope as function of time, after seeding one neuron with an electrical potential (black line). The activity was measured by changes in calcium ion concentrations. These concentrations were detected by imaging fluorescence levels relative to the average fluorescence of the neurons (activity 0) measured prior to activation. In the sparse cultures with few synapses per neuron, the stimulated neuron evokes a network burst of activity in all other neurons in the field after a short delay. By contrast, in the dense cultures with many synapses per neuron, only the stimulated neuron has an increased potential. The data for these plots were kindly provided by Ivenshitz & Segal [41]. (a) Low connectivity and (b) high connectivity.*

## Minor tips

1. Find and read a copy of an English writing style guide for academic writing. A famous one is the (old) Elements of Style by William Strunk, e.g. https://faculty.washington.edu/heagerty/Courses/b572/public/StrunkWhite.pdf .
    1. For instance, do you know how to write out an enumeration? Mind the second comma: "I

   did this, that, and more." If your enumerated phrases contain comma's (which is rare) then use semi-colon as separator instead. Also do that if you make your enumeration into a bulleted/numbered list.
   2. Pick either US or British English and stick with it. Usually your envisioned journal will have a preference, so look that up.
2. Keep your English as simple and to the point as possible. Most readers will be non-native English speakers so do not show off your knowledge of rare and complicated words, nor use phrases like 'en masse' (originally French) which not every non-native English speaker understands.
3. Use the active form for sentences ("We study...") instead of inactive ("What has been studied is...").
4. Use concise sentences. Do not compound many facts or considerations into a single long sentence with two or three commas.
5. Equations on separate lines (possibly with an equation number) are still part of the text. So use proper punctuation, most often a comma or a period as last character of the equation, so that the line immediately after either continues with the same sentence (and lower case) or a new sentence, respectively.

### Further reading / external links

1. So you're writing a paper
2. 'Scientific Papers' by Nature.